

ICST INETRDISCIPLINARY CENTER FOR SMART TECHNOLOGIES
Center of CS Department's Software Engineering Track

SmartRail

Spread out. Board faster.

The problem: passengers bunch near the entrances, so a few doors take most boarders and hold the train.

SmartRail senses free train seats and shows them on the platform, so passengers spread out before the train arrives.

- **Seat sensing:** ESP32 + IR sensor above each seat
- **Platform map:** each car's free seats at its zone
- **Phone app:** a different car for each passenger

Result: in simulation, stops are 22% shorter.

Seat sensors
IR sensor above each seat
2 real seats: A1, B1

→ analog

ESP32
1 Calibrate
2 Filter and decide
3 Send seat states

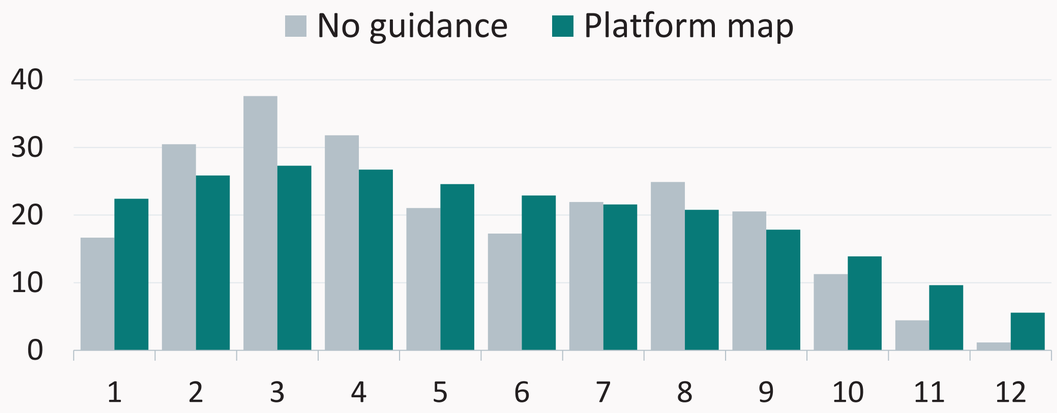
→ USB serial

Laptop server
4 Add 1,294 simulated seats
5 Count free seats per car

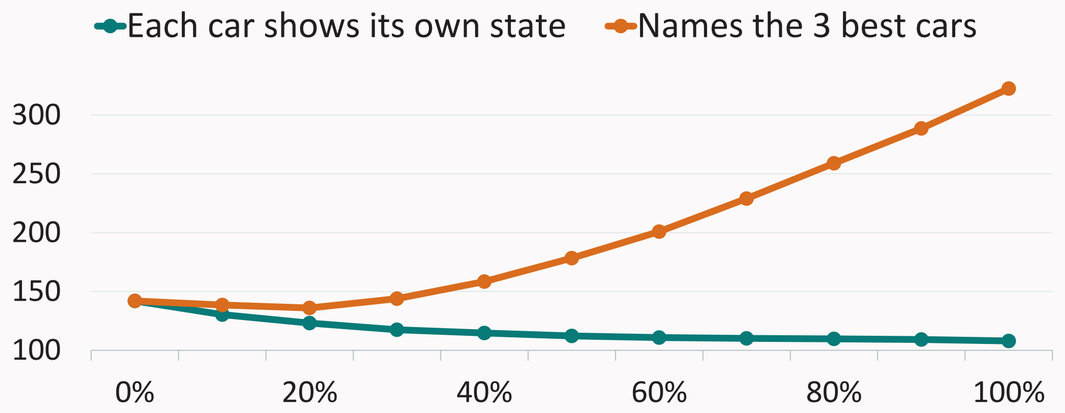
→ Wi-Fi / LAN

Passengers
6 Platform map + phone app

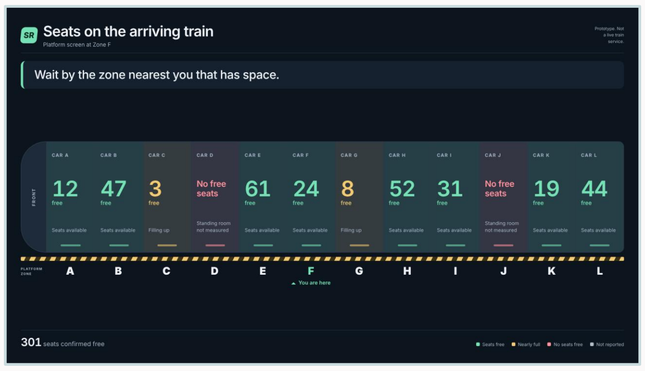
System diagram: from seat sensor to the passenger, steps 1–6



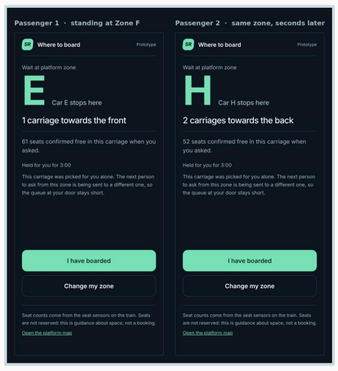
Boarders per car: the crowd spreads out



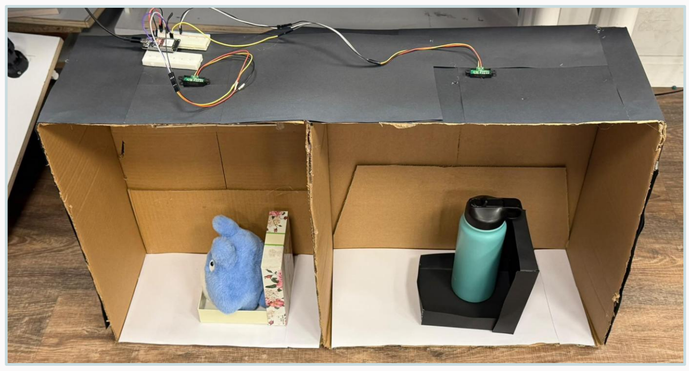
Stop time (s) vs. share who follow it



Platform screen: no ranking



Phone app: one car each



Seat sensor prototype

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A Project in Internet of Things (IoT)

Computer Science Department
הטכניון מכון טכנולוגי לישראל